

PART B

Splash & Swim Club is a local organization that manages annual swimming competitions for schools and independent participants. The number of participants has increased significantly in recent years, and the club has decided to develop a database system to record and manage all competition-related information efficiently. Below are the requirements and business rules provided by the club:

- The system must keep track of swimmer information, including swimmer ID, name, gender, date of birth, contact number, and representing school or club name.
 - Details of events must be recorded, including event ID, event name (e.g. 100m Freestyle, 50m Butterfly), and event category (Men/Women).
 - Each swimmer can register multiple events in one competition.
 - Each event can be in multiple registration, which the registration process is supervised by exactly one official.
 - An official also may supervise many registrations process.
 - The system must also record official information such as official ID, name, role (e.g. referee, timekeeper, starter), and contact number.
 - Swimmers' performance for each registered event, should be recorded which include the name of the round (e.g heat 1, quarter final, final), date of the round, competition time (in seconds) and position (rank).
 - The system should also store venue information, such as venue ID, venue name, location, and the pool size (e.g. 25m or 50m pool).
 - Every competition takes place at one venue, but a venue can host multiple competitions over time.
- b) If an official cannot supervise more than 100 registered events, by using Chen notation, show the full diagram between official and the registration. (3.5 marks)